

A Finite Margin-Stability Matrix for the Literal V59 Residual

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Abstract

TPC-272 identified the correlation margin $m = |C_{\perp}|/R$ as a necessary input for converting a signed residual estimate into an endpoint estimate. We now test whether that margin is stable under the declared finite cutoff, clock, and kernel choices, while keeping the literal V59 physical operator fixed. On a 32-row grid consisting of four scales, four comparison cutoffs, and two kernel exponents, outward rational intervals classify 12 rows with $m < 1/8$, 11 middle-band rows, and 9 rows with $m > 1/4$. At fixed scale and exponent, changing only the cutoff crosses from the middle band to the high band at $N = 64$, and from the middle band to the low band at $N = 128$. The same grid has 30 negative-real and two positive-real scalar phases. This is a certified finite stability obstruction for the declared interface, not an asymptotic counterexample, a source-level theorem, or a twin-prime proof.

1 Question and claim ceiling

The residual route has a signed scalar $C_{\perp}$ and a radius $R = \sqrt{W_{\perp}G_{\perp}}$. TPC-272 proved that a scalar saving and a quantitative lower bound on $m = |C_{\perp}|/R$ combine as a $\sigma - \eta$ endpoint budget. A finite phase label is not such a lower bound. The present paper makes the next minimal hostile test: hold the physical operator, masks, shell, beta, and projection fixed, and vary only declared local cutoff and kernel parameters.

The claim ceiling is

NUMERICALLY_CERTIFIED_FINITE, REFUTED_SCOPED,

for the registered matrix. “Scoped” is part of the statement: the grid does not represent a proved growing sequence. Fixed-power credit, arithmetic L^2 , full Gate B, and the twin-prime conclusion remain absent.

2 Frozen finite object

For each N we use the same finite literal operator as the parent TPC-268 audit: prime shell $Q < q \leq 2Q$, outer weight q , the two unit masks, deleted diagonal, exact finite beta, and the four consecutive equal blocks with their three Haar contrasts. The physical scale triples are

$$(N, H, Q) = (64, 15, 4), (96, 20, 5), (128, 24, 5), (192, 32, 6).$$

For a cutoff $z \in \{2, 3, 4, 5\}$ and exponent $s \in \{1, 2\}$, the kernel is $K_H(h) = (1 + (h/H)^2)^{-s}$. No source coefficient is replaced by a synthetic random vector.

Let P_3 be the declared rank-three block projection. The parent engine returns the interval for

$$\rho^2 = \frac{|C_\perp|^2}{\|(I - P_3)w\|^2 \|(I - P_3)g\|^2}.$$

The residual norm intervals are positive and the scalar interval is separated from zero on every registered row.

Lemma 1 (margin transfer). *On every registered row,*

$$m^2 = \rho^2, \quad m^6 = (\rho^2)^3.$$

Proof. Both quantities are the same quotient after substituting $R^2 = \|(I - P_3)w\|^2 \|(I - P_3)g\|^2$. Positivity makes squaring and cubing monotone, so the parent outward interval transfers directly. $\square$

This elementary identity is useful operationally: the certificate never extracts a floating-point square root to decide a margin band.

3 The 32-row matrix

The low and high tests are deliberately separated:

$$\rho^2 < \frac{1}{64} \implies m < \frac{1}{8}, \quad \rho^2 > \frac{1}{16} \implies m > \frac{1}{4}.$$

Rows failing both tests are retained as the middle band. The resulting counts are

	$m < 1/8$	$1/8 \leq m \leq 1/4$	$m > 1/4$
rows	12	11	9.

The exact JSON stores rational endpoints; a few representative intervals are shown below in decimal form.

N	z	s	interval for m^2	phase
64	2	1	[0.0538214, 0.0538299]	negative
64	5	1	[0.1482570, 0.1483212]	negative
96	3	1	[0.0898035, 0.0898164]	negative
128	3	1	[0.0155789, 0.0155831]	negative
192	3	1	[0.00010335, 0.00010354]	positive
192	5	1	[0.00005266, 0.00005285]	negative

Table 1: Representative outward intervals. The complete matrix has 32 rows.

Theorem 1 (finite margin-stability obstruction). *The 32 registered rows have 12 low, 11 middle, and 9 high margin classifications. At fixed $(N, s) = (64, 1)$, the cutoff-only transition $z = 2 \rightarrow 5$ is middle-to-high. At fixed $(N, s) = (128, 1)$, the cutoff-only transition $z = 2 \rightarrow 3$ is middle-to-low. Each transition is separated by its stored outward intervals. Therefore uniform margin stability for this declared finite parameter family is **REFUTED_SCOPED**.*

Proof. The producer evaluates all 32 cases and applies the two rational threshold tests above. The independent checker recomputes each row through the frozen parent engine and transfers the interval using Lemma 1. For $N = 64$, the $z = 2$ upper endpoint is below $1/16$, whereas the $z = 5$ lower endpoint is above $1/16$. For $N = 128$, the $z = 2$ row lies above $1/64$, whereas the $z = 3$ upper endpoint lies below $1/64$. Since only the declared cutoff changes in each comparison, the asserted finite instability follows. $\square$

The corresponding squared-margin ratios are approximately 2.75 and 0.969, respectively. They are finite transition diagnostics, not exponents in N .

4 Phase census and kernel control

The scalar phase census contains 30 `NEGATIVE_REAL_AXIS` rows and two `POSITIVE_REAL_AXIS` rows. The positive rows are exactly

$$(N, z, s) = (192, 3, 1), \quad (192, 4, 1).$$

They are retained in the matrix because deleting phase flips would make the stability test circular. At $N = 96, z = 3$, changing only $s = 1 \rightarrow 2$ keeps both rows in the high margin band; the associated squared-margin ratio is enclosed near 0.928. Thus the present obstruction is specifically driven by cutoff sensitivity in the two displayed transitions, while the kernel control provides a matched non-flip.

Theorem 2 (finite phase census). *Within the declared grid, the phase counts are 30 negative-real, 2 positive-real, and 0 zero-crossing intervals. This census does not imply an eventual phase sector and does not imply a positive margin lower bound.*

Proof. The scalar intervals supplied by each parent-engine evaluation are separated from zero. Counting their signs gives the stated numbers; the independent replay checks the same keys and signs. The final sentence follows from the two-dimensional sign-only construction already formalized in TPC-272, and is also visible here from the coexistence of low-margin rows with either sign. $\square$

5 Interpretation and limits

The new paper makes one narrow advancement: it converts the qualitative “cutoff may matter” warning into a threshold-separated margin matrix. It does not claim that the finite cutoff values are the source-level growing rule, nor that a finite crossing supplies an asymptotic subsequence. In particular,

1. no row count is converted into a power saving or loss;
2. the two positive phase rows are not discarded or averaged away;
3. the parent operator and its exact beta are unchanged;
4. arithmetic L^2 , signed four-packet reassembly, and full Gate B are untouched.

The next legitimate source-level question is uniformity on the registered growing cutoff itself. If that theorem fails, its failure must be stated in the actual source family rather than inferred from this finite matrix.

6 Conclusion

TPC-273 supplies a hostile finite test of the new TPC-272 margin coordinate. The same literal finite operator exhibits both low and high correlation bands under cutoff-only changes, and it also exhibits two phase sign flips. The result is a useful obstruction to treating a declared finite interface as automatically stable. It leaves the source-level margin lower bound, the strict $1/400$ payment, arithmetic L^2 , and the twin-prime problem open.

References

- [1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.
- [2] L. Wang, “A Correlation-Margin Budget for the V59 Residual,” TPC-272 project release, 2026.